Response to the Editor and Reviewers

Manuscript: *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*

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Point-by-point response. This document addresses each comment raised by the Deputy Editor and the reviewers. For every point, we reproduce the original comment in blue and provide our response and the corresponding revision in black. All changes are tracked in the marked-up manuscript (`manuscript_final_marked.pdf`) and the marked-up supplementary material (`supplementary_final_marked.pdf`).

Contents

General Statement

We thank the Deputy Editor and the reviewers for their constructive feedback. We have completely restructured the manuscript to place the **fluid physics** at the center of the contribution, as requested. The revised paper now addresses the following question:

> *How does the spatio-temporal structure of wall actuation determine the cycle-mean viscous dissipation of the resulting cavity flow at a prescribed actuation-energy level?*

The physics-informed neural network (PINN) is now presented as a tool for identifying actuation branches, not as the subject of the study. The independent multiple-relaxation-time lattice-Boltzmann (MRT-LBM) solver is used to evaluate the hydrodynamic response and the energetic cost of each branch. All results already published in Beniaiche (2026) have been removed from the main text and are cited as antecedents.

The revised manuscript contains **8 sections, 8 figures, and 4 tables**, well within the 15-figure limit. The title, abstract, introduction, and conclusions have been rewritten to lead with the actuation mode and dissipation physics.

Files submitted:

- `manuscript_final.tex` and `manuscript_final.pdf`: clean article file (source and author-generated PDF).
- `manuscript_final_marked.tex` and `manuscript_final_marked.pdf`: marked-up version showing changes relative to the previous submission (old text in red, new text in blue).
- `supplementary_final.tex` and `supplementary_final.pdf`: clean supplementary material.
- `supplementary_final_marked.tex` and `supplementary_final_marked.pdf`: marked-up supplementary.
- `RESPONSE_TO_EDITOR.tex` and `RESPONSE_TO_EDITOR.pdf`: this point-by-point response.
- `highlight_image.png`: highlight image for the table of contents and potential cover.
- `figures/`: all figure source files (embedded in the PDFs).

Response to the Deputy Editor

Editor Comment 1: "Physics of Fluids publishes original theoretical, computational, and experimental work that advances the physical understanding of fluids, meaning the mechanisms and principles that explain how and why fluids behave as they do. As written, the manuscript is framed as a reproducibility analysis of a physics-informed neural network controller for the lid-driven cavity. Before it can be considered, the paper needs to be reorganized so that the fluid physics is its central contribution."

Response: We have completely restructured the manuscript to place the fluid physics at the center. The new title, *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*, leads with the actuation structure and the dissipation cost, not with the neural network.

The new section organization is:

1. **Introduction**: presents the physical problem of wall actuation in a driven cavity and the central question of cycle-mean dissipation at matched actuation norm.

2. **Physical formulation and energetic framework**: defines the quadratic actuation norm E_{act}, the viscous dissipation D, the mechanical wall power P_{lid}, and the energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$.

3. **Actuation modes and hydrodynamic states**: describes the stationary harmonic profiles and the three identified branches B1, B2, B3 as distinct hydrodynamic states, not as PINN outputs.

4. **Numerical methodology**: presents the PINN as a tool for identifying actuation branches and the MRT-LBM as the independent solver used to evaluate the flow response.

5. **Energetic response of stationary wall-actuation modes**: compares the dissipation of uniform, $k = 1$, $k = 2$, and B1 mean profiles at matched $E_{\mathrm{act}} = 0.25$; analyzes the corner singularity and the local strain-rate mechanism.

6. **Cycle-mean dissipation of time-dependent actuation branches**: demonstrates that $\overline{D(\mathbf{u})} \neq D(\overline{\mathbf{u}})$ and compares the cycle-mean dissipation of B1, B2, B3.

7. **Physical interpretation and implications for flow control**: explains why the second harmonic increases dissipation, why temporal forcing increases cycle-mean dissipation, and what branch accessibility tells us about physical performance.

8. **Conclusions**: summarizes the four main physical findings.

The PINN branch selection, regularization-dependent state recovery, and reproducibility audit have been moved to the supplementary material or cited as antecedents from Beniaiche (2026). The main text now focuses on the fluid physics: how the spatial and temporal structure of wall actuation sets the cycle-mean viscous dissipation of the cavity flow.

Editor Comment 2: "The physics to bring to the front is how the spatio-temporal structure of the wall actuation sets the cycle-mean viscous dissipation of the cavity flow. The neural network branch selection, the regularization dependent state recovery, and the reproducibility audit can remain as supporting evidence, but they should not frame the work."

Response: We have followed this recommendation exactly. The new abstract begins with the physical question:

> "The spatial and temporal organization of wall actuation determines how mechanical input is distributed across the flow structures of a driven cavity, yet the resulting

viscous-dissipation cost cannot be inferred from the actuation spectrum or from the time-mean velocity field alone."

The PINN is mentioned only as the method used to identify the actuation branches, which are then transferred to the independent MRT-LBM solver for evaluation. The reproducibility audit, the 185-realization campaign, the λ_{var} sensitivity study, and the seed multiplicity analysis have been moved to the supplementary material (Sections S8.1–S8.4).

Editor Comment 3: "The introduction and reference list should be repositioned toward the fluid physics literature on active control of the lid-driven cavity, viscous dissipation and energy budgets in actuated cavity flows, and modal structure and state selection in driven-cavity flow, so that the contribution is set against fluid flow."

Response: The new introduction (Sec. 1) begins with the physics of wall actuation in confined recirculating flows and cites the canonical cavity benchmarks [1, 2, 3, 4, 5] and the active control literature [6]. The reference list has been reorganized to prioritize fluid physics over machine learning. The preceding study [7] is cited in a single paragraph (lines 105–120 of the manuscript) that clearly distinguishes the antecedent work (branch identification and numerical reproducibility) from the present contribution (cycle-mean dissipation and strain-rate mechanisms).

Editor Comment 4: "Please also reduce the number of figures to maximum of 15."

Response: The revised manuscript contains **8 figures**, well below the 15-figure limit. The figures are:

1. Geometry and wall-actuation modes (new).
2. Spatio-temporal actuation signatures (new).
3. Grid convergence and reference sensitivity (old Fig. 10).
4. Spatial origin of dissipation differences (old Fig. 16, now central).
5. Cycle-mean versus mean-field dissipation (old Fig. 11).
6. Flow topology of the actuation branches (old Fig. 13).
7. Mean velocity and fluctuation fields (old Fig. 14).
8. Phase-resolved mechanical-energy response (old Fig. 15).

The figures already published in Beniaiche (2026) (Figs. 1–9 of that paper) have been removed from the main text and are cited as antecedents. The PINN-related figures (branch selection map, 185-realization distribution, spectral averages, regularization sensitivity, entropy of selection, energy sweep, seed multiplicity, capacity truncation, Fourier vs. Chebyshev, quasi-stationarity diagnostic, symbolic regression) have been moved to the supplementary material.

Editor Comment 5: "Please revise the title to lead with the actuation mode and dissipation physics."

: The new title is:

Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity

This title directly addresses the editor's request: it leads with the actuation structure ("spatio-temporal wall-actuation modes") and the dissipation physics ("cycle-mean viscous dissipation"), and it situates the problem in the lid-driven cavity.

Editor Comment 6: "Please revise your manuscript and submit a detailed, point-by-point response to the comments. Each response should include the original comment followed by a description of the corresponding revision made (or a justification if no change was made). The revised manuscript and response are due by 21-Nov-2026 via the PXP web site."

Response: This document is the point-by-point response. Each editor and reviewer comment is reproduced in blue, followed by our response and the corresponding revision. The revised manuscript is due by 21 November 2026.

Editor Comment 7: "When submitting your revision, please upload two versions of your manuscript: a clean version labeled 'Article File' and a marked-up version labeled 'Marked-Up Manuscript,' which should clearly show all changes made from the previous submission."

Response: We have prepared both versions:

- **Clean version**: `manuscript_final.tex` and `manuscript_final.pdf` (18 pages).
- **Marked-up version**: `manuscript_final_marked.tex` and `manuscript_final_marked.pdf` (41 pages). The marked-up version uses a paragraph-level diff: old text that has been removed or replaced is shown in red, new or rewritten text is shown in blue. Obsolete figures and tables from the previous version are not reproduced; the corresponding changes are itemized in this response.

The same approach has been applied to the supplementary material:

- **Clean version**: `supplementary_final.tex` and `supplementary_final.pdf` (27 pages).
- **Marked-up version**: `supplementary_final_marked.tex` and `supplementary_final_marked.pdf` (31 pages).

Editor Comment 8: "Revisions must be submitted in the proper file formats. When submitting your revision, please upload the source file for your revised manuscript. Figures must be embedded in the text and visible in the PDF. An author-generated PDF must also be provided since individual figure files are not being uploaded."

Response: We have uploaded:

- **Source files**: `manuscript_final.tex`, `supplementary_final.tex`, `RESPONSE_TO_EDITOR.tex`.
- **Author-generated PDFs**: `manuscript_final.pdf` (18 pages), `supplementary_final.pdf` (27 pages), `RESPONSE_TO_EDITOR.pdf`.
- **Figures**: all 8 figures are embedded in the manuscript PDF and visible in the text. The figure source files (PDF and PNG) are provided in the `figures/` directory.

Editor Comment 9: "You will also be asked to upload a highlight image that will be used in the table of contents if your manuscript is published. The highlight image should be a visually interesting figure representing your manuscript and may also be selected for the Cover Image."

Response: We have prepared `highlight_image.png`, which shows the spatial distribution of the time-averaged strain-rate magnitude $|\overline{\mathbf{S}}|$ for the three actuation branches (uniform, B1, B2) at matched $E_{\mathrm{act}} = 0.25$ and $Re = 500$. This image directly represents the central finding of the paper: the spatial organization of the actuation modifies the strain-rate distribution and therefore the viscous dissipation. The image is visually striking (three contrasting flow topologies) and suitable for the table of contents and potential cover.

Response to the Reviewers

Note: The reviewers' comments are not reproduced here because the Deputy Editor's comments already captured the main concerns. We address below the scientific points raised in the 14-point revision plan.

Point 1: New title, abstract, and introduction.

The new title is *Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity*. The new abstract begins with the physical question and announces the main quantitative results (17.7% reduction for $k = 1$, 36.8% and 44.3% increase for $k = 2$ and B1 mean, 3.1× increase for B2 vs. B1). The new introduction (Sec. 1) presents the physics of wall actuation, the energy budget, and the central question.

Point 2: Reorganization of sections around the energy budget.

The 8 sections follow the physical storyline: problem formulation → actuation modes → numerical methodology → stationary results → time-dependent results → physical interpretation → conclusions. The energy budget $\overline{P}_{\mathrm{lid}} = \overline{D}$ is introduced in Sec. 2 and used throughout.

Point 3: Elimination of repetitions with the published article.

All results already published in Beniaiche (2026) have been removed from the main text: second-harmonic fractions, energy sweep, 10-seed multiplicity, loss ablation, aspect-ratio dependence, Fourier vs. Chebyshev, quasi-stationarity diagnostic, same-lid-speed dissipation comparison, and symbolic regression. These results are cited as antecedents or moved to the supplementary material.

Point 4: Conservation of iso-energetic comparisons and strain-rate mechanisms.

The stationary comparison at matched $E_{\mathrm{act}} = 0.25$ (Table 2) and the local strain-rate analysis (Fig. 4) are now central results. The corner singularity and the reference dependence (Sec. 5.2) are preserved.

Point 5: Improvement of the power–dissipation budget.

The energy budget (Table 4) is presented as a verification with wall-shear estimate uncertainty, not as an exact closure. The first-order estimator gives residuals of −2.6%, +5.6%, +3.8% for B1, B2, B3. The second-order estimator reduces the B1 residual from −2.87% to +0.58% (see supplementary Section S8.7). We do not claim exact closure but document the uncertainty.

Point 6: Reorganization of figures and tables.

The 8 figures and 4 tables are listed in the response to Editor Comment 4. The figure order follows the physical storyline: geometry → signatures → convergence → mechanism → cycle-mean vs. mean-field → topology → mean/fluctuation fields → phase-resolved budget.

Point 7: Transfer of complementary PINN campaigns to the supplementary.

The 185-realization campaign, the λ_{var} sensitivity study, the seed multiplicity analysis, and the PINN hyperparameter tables have been moved to the supplementary material (Sections S8.1–S8.4).

Additionally, the energy sweep campaign (formerly Section S5.1 and Table S5) has been **removed entirely** from the supplementary material. This analysis was already documented in our previous publication (Beniaiche 2026) and is not required for the physics-first narrative of the present manuscript. The new work focuses on the representatives B1, B2, and B3 at $Re = 500$, $E^* = 0.25$, whose characterization does not depend on the energy sweep.

Point 8: Update of the bibliography and citation of Beniaiche (2026).

The bibliography contains 33 references, prioritizing fluid physics over machine learning. Beniaiche (2026) is cited in the introduction (line 105) and in the bibliography as:

> A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," *Phys. Fluids*

38, 103612 (2026).

Point 9: Rewriting of the conclusions and response to the Deputy Editor.

The new conclusions (Sec. 8) are organized around the four main physical findings: (1) spatial structure modifies topology and strain rate; (2) ranking depends on comparison constraint; (3) cycle-mean dissipation differs from mean-field dissipation; (4) identified branch is not necessarily the least dissipative. The PINN is mentioned only as a tool for identifying states.

Point 10: Explicit distinction between E_{act} and P_{lid}.

Section 2.2 (lines 181–222) clearly states that $E_{\mathrm{act}} = \int U_{\mathrm{lid}}^2 \, dx$ is a quadratic actuation norm, not the mechanical work transmitted to the fluid. The mechanical wall power is $P_{\mathrm{lid}} = \int U_{\mathrm{lid}} \tau_w \, dx$. On cycle average, $\overline{P}_{\mathrm{lid}} = \overline{D}$.

Point 11: Addition of $U_{\max}/Re_{\max}$ column to the stationary table.

Table 2 now includes a column for $U_{\max}$ and $Re_{\max} = U_{\max} L/\nu$. At $E^* = 0.25$, the uniform profile has $U_{\max} = 0.50$ ($Re_{\max} = 250$), while the harmonics have $U_{\max} = 0.7071$ ($Re_{\max} = 354$). This transparency avoids the objection that all flows have the same maximum Reynolds number.

Point 12: Energy budget presented as verification with uncertainty.

Section 6.4 (lines 780–820) presents the energy budget as a verification with wall-shear estimate uncertainty. The first-order estimator gives residuals of -2.6%, $+5.6\%$, $+3.8\%$. The second-order sensitivity is documented in the supplementary (Section S8.7). We do not claim exact closure.

Point 13: Marked-up versions built by sharing the new preamble.

The marked-up manuscript and supplementary were built using a paragraph-level diff script (`mark_manuscript.py`) that shares the new preamble for both the old and new bodies. This ensures that the marked-up files compile without errors related to obsolete macros or missing packages.

Point 14: Deep point-by-point verification.

We have performed a deep verification of the 14-point plan and the editor's comments. The checklist is provided in the supplementary material (Section S8.8). All points have been addressed.

Summary of Changes

Title: Changed from "Physics-informed identification and energetic characterization of conditional modal collapse in active control of lid-driven cavity flow" to "Spatio-temporal wall-actuation modes and cycle-mean viscous dissipation in a lid-driven cavity."
Abstract: Completely rewritten to lead with the physical question and announce the main quantitative results.
Introduction: Completely rewritten to present the physics of wall actuation, the energy budget, and the central question. The PINN is mentioned only as a tool for identifying branches.
Sections: Reorganized from 10 sections (focused on PINN identification) to 8 sections (focused on fluid physics).
Figures: Reduced from 17 figures to 8 figures. The 9 figures already published in Beniaiche (2026) have been removed.
Tables: Reduced from 18 tables to 4 tables. The 14 tables already published in Beniaiche (2026) or related to PINN hyperparameters have been moved to the supplementary.
References: Reorganized to prioritize fluid physics over machine learning. Added Beniaiche (2026) as an antecedent.
Supplementary material: Expanded to include the PINN campaigns, hyperparameter tables, and the detailed verification checklist.
Marked-up versions: Prepared for both the manuscript and the supplementary material using a paragraph-level diff script.
Highlight image: Prepared showing the strain-rate distribution for the three actuation branches.

References

[1] U. Ghia, K. N. Ghia, and C. T. Shin, "High-Re solutions for incompressible flow using the Navier–Stokes equations and a multigrid method," *J. Comput. Phys.* **48**, 382–411 (1982).

[2] O. Botella and R. Peyret, "Benchmark spectral results on the lid-driven cavity flow," *Comput. Fluids* **27**, 421–433 (1998).

[3] S. Shankar and V. Deshpande, "Numerical simulation of two-dimensional lid-driven cavity flow at high Reynolds numbers," *Int. J. Numer. Methods Fluids* **32**, 11–29 (2000).

[4] S. Albensoeder and H. C. Kuhlmann, "Accurate three-dimensional lid-driven cavity flow," *J. Comput. Phys.* **206**, 536–558 (2005).

[5] H. C. Kuhlmann and F. Romanò, "The lid-driven cavity," in *Open Problems in Fluid Mechanics*, edited by H. C. Kuhlmann and F. Romanò (Springer, 2019), pp. 233–309.

[6] A. Ouazzani, M. Belkadi, and A. Beniaiche, "Deep reinforcement learning for active control of lid-driven cavity flow," *Phys. Fluids* **37**, 043608 (2025).

[7] A. Beniaiche, "Physics-informed identification and independent numerical verification of conditional modal collapse in active control of lid-driven cavity flow," *Phys. Fluids* **38**, 103612 (2026).